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REPRINT: THE HISTORY, PHILOSOPHY AND DEVELOPMENT OF COMPUTATIONAL INTELLIGENCE (HOW A SIMPLE TUNE BECAME A MONSTER HIT)[1]

James C. Bezdek
Computer Science, U. of Melbourne, Parkville, Vic., Australia

Keywords: Computational Intelligence, Artificial Intelligence, Biological Intelligence, Neural Networks, Fuzzy Systems, Evolutionary Computation

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Summary

Hisao Ishibuchi asked me to write the first chapter in this volume. He suggested the title "History, Philosophy and Development of Computational Intelligence" but for reasons that will soon be clear, I added the subtitle you see above. Why ask me to write it? Well, [I think] I wrote the first paper that defined the term computational intelligence (Bezdek, 1992), and it was my idea to attach the term computational intelligence to several activities related to the *IEEE Computational Intelligence Society* (IEEE CIS). The most important event in this regard is the World Congress on Computational Intelligence (WCCI) first held in Orlando in 1994. Indeed, the current name of the society – IEEE CIS – also had its roots in my suggestion. But, as I have been careful to point out many times, the term CI itself was around for at least seven years before I wrote that paper. This article finally sets that part of the history straight. Let me start this excursion into the past with a seemingly unrelated but soon to be understood word association game about popular music.

1. Prelude: Art and Science share a common trait

Suppose I name the song "Like a Rolling Stone" which the magazine *Rolling Stone* lists as the greatest song of all time. (You may not agree that this is the greatest song of all

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[1] James C. Bezdek, 2013, The History, Philosophy and Development of Computational Intelligence (How a simple tune became a Monster hit), in *Computational Intelligence*, edited by Hisao Ishibuci, Eolss book chapter 6.44.40.1, pp. 1-22, in Encyclopedia of Life Support Systems (EOLSS), Developed under the Auspices of the UNESCO, Eolss Publishers, Oxford, UK, [http://www.eolss.net]

time, or even of any time, because of its context, language, your cultural history, your personal preferences and so on – that's ok, it will still suffice to make my point.) Many of you will know who wrote it, who performed it first, and who made it a huge hit; Bob Dylan (1965) in all three cases. But what about the song "[I'm Dreaming of a] White Christmas"? I'm pretty sure that many of you can tell me who made it popular (Bing Crosby), but probably many of you don't know that Irving Berlin wrote it in 1940, and that Bing did make the first, and also most popular, recording of it in 1942[2]. You may be surprised to learn that the first hit of one of America's most distinguished "pop" singers, Tony Bennett, was Hank Williams' (arguably best) country song, "Cold, Cold Heart."

It is often the case in music that the general public attributes the creation of a well-known song to the artist who made it most popular – the song writer is often obscured by the dazzling success of the performer who makes it a big hit. There are lots of examples in music of anomalies, quirks, little-known facts, obscure references, "guest" credits for songwriters, and general mis (or is it dis?) information about songs. This happens in science and engineering too.

2. Overture: Songwriters and Performers in Science and Engineering

Suppose I state a phrase, say a technical term, name a concept, or repeat a physical law, that is common in science or engineering, and ask you to associate a name with that phrase, term, concept or law. For example, the law "energy equals mass times the square of the speed of light"; in symbols, $E = mc^2$. Almost everyone on planet earth (well, ok, almost everyone amongst the more than 400,000 members of the IEEE anyway) can tell me that this equation was discovered (created), and popularized, by Albert Einstein.

But suppose I ask for the origin of the term "*genetic algorithms* (GAs)"? What name leaps into your mind first? Perhaps David Goldberg, who wrote the first popular text about this topic (Goldberg, 1989). Or your response might be John Holland, who is sometimes credited as the "inventor" of GAs (Holland, 1975). The actual history of genetic algorithms is quite complicated, and the origins of various algorithmic forms of evolutionary computation can be rightfully attributed to several creators, but certainly not to Goldberg. See Fogel (1998) for an eminently readable and cogent account of the actual history of GAs as well as other forms of evolutionary algorithms.

Here's another example: "backpropagation in multilayered neural networks". This very famous and useful technique was created and reported by Paul Werbos in his PhD thesis (Werbos, 1974), but it was, for many years, erroneously credited to David Rumelhart and James McClelland, who popularized it in their 1986 book (Rumelhart and McClelland, 1986).

Where does all this lead? Suppose I ask the membership of the IEEE: Who originated the term "*computational intelligence* (CI)"? Many – perhaps most - of them might say "Jim Bezdek" but they would be wrong. I probably made this term a big hit (song), but I did

---

[2] According to the Guiness book of World Records, the holiday perennial "White Christmas" (1942) by Bing Crosby is the best-selling single worldwide, with estimated sales of over 50 million copies.

not write the song. That piece of the story will come to light soon. But first, let's return to the music analogy, which was not my invention either!

I will start by discussing the history of a different technical term whose path is strikingly similar to the historical evolution of the term CI. In particular, there is a close parallel between the terms CI and EM, which stands for "*expectation-maximization* (EM*)*." I have done a lot of work related to the theory of what some writers call *alternating optimization* (AO). AO is the scheme employed by EM when estimating the unknown parameters of a bunch of mixed probability distributions. In due course, I became very interested in trying to track down the history of AO, and my inquiries into this topic became somewhat inseparable from the history of the EM algorithm. Let me call this combined history EM/AO. It turns out that the history of various EM/AO algorithms is pretty cloudy. Several scholars have written quite interesting and rather comprehensive histories related to EM/AO. But the history of the *term* EM itself is pretty well known. Here is the opening section of (Meng and Van Dyk, 1997), reproduced in its entirety.

Quote Q1: The EM Algorithm - an Old Folk-Song Sung to a Fast New Tune (Meng, Van Dyk, 1997)[3]

> **1.1.** Who First Developed the EM Algorithm? With the ever-growing popularity of the EM algorithm, especially with its various deterministic and stochastic extensions (e.g. the data augmentation algorithm of Tanner and Wong (1987)), those of us who do research in this area find ourselves being asked more frequently the question who first developed the EM algorithm? Although it is easy for us to direct the inquirer to Dempster et al. (1977) where the term EM appeared for the first time, the question is really not easy to answer. In fact, the issue of the origin of the EM method was raised by several discussants of Dempster et al. (1977). For example, Hartley opened his contribution with
>
> 'I felt like the old minstrel who has been singing his song for 18 years and now finds, with considerable satisfaction, that his folklore is the theme of an overpowering symphony'
>
> Hartley's 'folk-song' analogy is indeed appropriate for describing the development of this powerful method. Just as a folk-song typically evolves many years before its tune is well recognized, various EM-type methods or ideas which precede Dempster et al. (1977), and in fact precede Hartley (1958) by many years, can be found in the literature. For instance, the earliest piece of the EM score traced by Dempster et al. (1977) is McKendrick (1926). If we are willing to make a broader connection, then a key identity underlying the EM algorithm can be traced back, as with many other popular statistical methods (e.g. the bootstrap), to the work of Fisher (1925), as pointed out by Efron in his discussion of Dempster et al. (1977). The folk-song analogy is

[3] I have not replicated the following references imbedded in Q1 in my bibliography: Tanner and Wong, McKendrick, Fisher, Efron, Sundberg, Baum, A. Martin-Lof and Stigler. Please see Meng and Van Dyk (1997) for these citations.

> also accurate in the sense that it signifies the collective effort in developing the EM algorithm. Indeed, a couple of dozen individuals were credited by Dempster et al. (1977) for contributing to one degree or another, some with new verses and some with remakes. Among them, Baum et al. (1970) is perhaps the most sophisticated -we still cannot sing it without first warming up with the version of Dempster et al. (1977). This is not a criticism of Baum et al. (1970), who might have been required by their publisher to adopt such a compact version, but merely a remark attempting to explain why their version, which had the key notes as did Dempster et al. (1977), did not become the hit that Dempster et al. (1977) did seven years later. Combining Baum et al. (1970) with Sundberg (1974, 1976), which were based on the author's thesis at Stockholm University (Sundberg, 1972), perhaps would have caught more attention. Sundberg not only provided an easily accessible rendition of the theory underlying the EM algorithm when the complete data are from an exponential family (where the algorithm is most useful) but also illustrated the iterative method with several examples. What was missing in Sundberg's version was an explicit result on the monotone convergence in likelihood, a celebrated feature of the EM algorithm, which was proved in Baum et al. (1970). As a further note on the difficulties in answering the question of the origin of the EM algorithm, Sundberg (1976) acknowledged that his key 'iteration mapping', which corresponds to the EM mapping defined by Dempster et al. (1977), was suggested by A. Martin-Lof in a personal communication.
>
> Although we shall perhaps never be able to find out who really sang the first musical note of the EM algorithm, we all agree that it was Dempster et al. (1977) who brought it into the all-time top 10 of statistics (see Stigler (1994)). They made (at least) two contributions that popularized the song. First, they gave it an informative title identifying the key stanzas-the expectation step and the maximization step. Second, they demonstrated how it could be sung at many different occasions, some of which had not previously been thought to be related to the EM algorithm (e.g. viewing latent variables as missing data). Since then, we all have sung or heard it being sung many times, sometimes with abusive or even unbearable tones.

So, as you can see, there is precedence in both the Arts and in the Sciences for this type of confusion, and it is no accident that there is a more or less direct analogue between the music analogy I made in the prelude to this article and the popularization of the terms EM and CI. The language of the EM history used by Meng and Van Dyk (1997) adapts remarkably well to the history of evolution of the term CI. Observe that CI is not quite semantically equivalent to EM, because EM refers to several AO algorithms, whereas CI is simply a broad-brush term that is used to describe – what? Well, that's the point of this article isn't it? Let's start by tracking down the beginning of this particular song.

3. Libretto: 1983 - Computational Intelligence Begins

The appearance of the term CI in published form goes back to at least 1983, for that is when the *Int. J. of Computational Intelligence* (IJCI) was floated as the title of a new Canadian journal by its founding editors, Nick Cercone and Gordon McCalla. Nick and Gordon both responded to my request for some information on their use of the term. Here is what each of them wrote to me in email communications:

Quote Q2: The Origin of CI as described by Nick Cercone (2012)

> Back in 1983 my colleague Gordon McCalla and I were the executives for the *Canadian Society for Computational Studies of Intelligence* (CSCSI), the oldest national AI society in the world, which began in 1974. We decided to start an AI journal to focus on pragmatic issues and AI systems and approached the Canadian National Research Council (NRC) which published journals. We decided that Computational Intelligence was a more fitting term than Artificial Intelligence after much debate; it seemed to describe our field more accurately. We thought AI was a bit of a misnomer. After satisfying their do-diligence and name searchers our NRC decided to publish IJCI. Gordon and I edited CI for 20 years and made the transition from NRC to Blackwell's (which bought it from NRC) which subsequently became Wiley-Blackwells. The logo is still the one drawn by a Quebec artist for the original NRC CI. The CSCSI has undergone a name change to CAIAC (Canadian Artificial Intelligence Association / Association pour l'intelligence artificielle au Canada). There have been many other journals and organizations since.

Subsequent to Nick Cercone's email to me, Gordon McCalla added the comments repeated in the next block of text:

Quote Q3: The Origin of CI as further described by Gordon McCalla (2012)

> Nick has the history down fairly well. The term "computational intelligence" was drawn from the name of our national AI society (Canadian Society for Computational Studies of Intelligence), which had been devised at the time of the society's founding around 1973-1974. We were further encouraged by Alan Mackworth, a well known computer vision and constraints scholar, who had used the term "computational vision" for the vision area way back in the 70's, and felt, as Nick has mentioned, that the name "computational intelligence" was a more appropriate name for our field than "artificial intelligence".
>
> The journal is still being published by Wiley-Blackwell, as Nick says, currently moving into volume 28. Its first issue appeared in February 1985, with four papers by: Candy Sidner (Plan parsing for intended response recognition in discourse); David Etherington, Robert Mercer, and Ray Reiter (On the adequacy of predicate circumscription for closed-world reasoning),

John Tsotsos (Knowledge organization and its role in representation and interpretation for time-varying data: the ALVEN system), and David Wilkins(Recovering from execution errors in SIPE).

So, you can see it covered a wide variety of AI: natural language, knowledge representation, vision, and planning. The entire first volume, similarly diverse, is available. Attach the editors' introduction to this first issue, which has some of the history of how the journal came to be. Subsequent volumes are also available from the Wiley site, and you can see the journal's evolution in content, some of which would currently still be called "computational intelligence", over the years (decades!!).

Finally, here are some excerpts from the editors' introduction to the first issue of IJCI:

Quote Q4: From *The Int. J. of Computational Intelligence* (IJCI, 1985)

Gordon McCalla and Nick Cercone, ***Computational Intelligence's*** first editors, will introduce the Journal's publisher and will explain the process through which your papers pass between submission and appearance in print. The current editors conceived of ***Computational Intelligence*** during late 1983. We suspected that though artificial intelligence (AI) papers were being published in journals specialized within subfields of AI, there exists a need for another comprehensive AI journal. Since 1970 the Journal ***Artificial Intelligence*** has been the only journal which reports results from the full spectrum of AI research. Research from the very large and expanding field of **AI** has been increasingly reported in conference proceedings, technical reports, and by personal communication. Research results were thus distributed as rapidly, but not as widely, as possible.

During late 1983 when the editors approached the National Research Council of Canada (NRCC) with the proposal for ***Computational Intelligence,*** the NRCC suggested that we consult our peers to determine whether they shared our interest in an additional general AI journal. Two hundred AI practitioners in Canada and abroad were surveyed. The more than 100 respondents practically unanimously agreed that many researchers do want to reach a general AI audience, and that they would probably be regular ***Computational Intelligence*** contributors. They agreed that the recent increase in popular interest in AI can best be informed by a general AI journal. The Canadian Society for the Computational Studies of Intelligence (Societt canadienne pour I'ttude de I'intelligence par ordinateur) agreed to sponsor ***Computational Intelligence.*** The CSCSUSCEIO is the longest established Canadian AI research association, including in its membership AI specialists of every subfield pursuing their interests in publicly and privately funded facilities. The NRCC considered the results of our survey and its own investigation and, in late June 1984, agreed to publish ***Computational Intelligence,*** its 13th journal in more than 50 years of distinguished scientific publishing.

> The title “Computational Intelligence” was chosen to reflect the fact that AI is distinct from other studies of intelligence in its emphasis on computational models. The name also seems short enough to be catchy, and general enough to attract submissions from all areas of AI.

For some historical perspective, you can contrast the contents of Q4 about CI with the definition of AI given in *Webster's New World Dictionary of Computer Terms (1988)* that was then current:

Definition (*Webster's New World Dictionary of Computer Terms (1988)*): Artificial Intelligence

> The branch of computer science that studies how smart a machine can be, which involves the capability of a device to perform functions normally associated with human intelligence, such as reasoning, learning, and self improvement, See EXPERT SYSTEMS, HEURISTIC, KNOWLEDGE BASED SYSTEMS, AND MACHINE LEARNING. /Abbreviated AI.

I think this section gives an accurate accounting for the beginning of the [published] term CI. Until I discover an earlier reference, I will take this as a correct description of the origin of this particular folk song. Things went along at a steady and quiet pace from 1983 to 1992. And then...

4. Aria: 1992 - The Horizon Expands

My first use of the term CI was in a paper that I wrote in 1991 that was published by the *Int. J. of Approximate Reasoning*, Bezdek (1992). Here is the abstract from that paper:

Quote Q5: Abstract (Bezdek, 1992)

> This paper concerns the relationship between neural-like computational networks, numerical pattern recognition and intelligence. Extensive research that proposes the use of neural models for a wide variety of applications has been conducted in the past few years. Sometimes the justification for investigating the potential of neural nets (NNs) is obvious. On the other hand, current enthusiasm for this approach has also led to the use of neural models when the apparent rationale for their use has been justified by what is best described as "feeding frenzy". In this latter instance there is at times a concomitant lack of concern about many "side issues" connected with algorithms (e.g., complexity, convergence, stability, robustness and performance validation) that need attention before any computational model becomes part of an operational system. These issues are examined with a view towards guessing how best to integrate and exploit the promise of the neural approach with other efforts aimed at advancing the art and science of pattern recognition and its applications in fielded systems in the next decade.

> A further purpose of the present paper is to characterize the notions of computational, artificial and biological intelligence; our hope is that a careful discussion of the relationship between systems that exhibit each of these properties will serve to guide rational expectations and development of models that exhibit or mimic "human behavior".

This paper was, to my knowledge, the first article that proposed a [somewhat loose] technical definition of the term CI. The meaning I intended for the term CI in this paper has been analyzed, re-analyzed, ridiculed, supported, criticized, lionized, and so on, almost ad infinitum (or should it be, ad nauseam?). For the record, here is Figure 1 from Bezdek (1992):

Input	Complexity →					Level
Sensory Data + Human Knowledge	BNN	⊂	BPR	⊂	BI	B ~ Organic
	∪		∪		∪	
Sensor Data + "Knowledge Tidbits"	ANN	⊂	APR	⊂	AI	A ~ Symbolic
	∪		∪		∪	
Sensor Data + Computation	CNN	⊂	CPR	⊂	CI	C ~ Numeric

Complexity ↑

Figure 1. The ABCs: Neural Networks, Pattern Recognition and Intelligence (Bezdek, 1992)

The abbreviations in Figure 1 are NN=Neural Network, PR=Pattern Recognition, I=Intelligence. Earlier in the paper I had posted my definition of the "ABCs":

A = *Artificial*	Non - Biological (Man-Made)
B = *Biological*	Physical + Chemical + (??) = Organic
C = *Computational*	Mathematics + Man-Made Machines

I am not going to regurgitate each and every point I was trying to make in 1992, but I will revisit several observations related to my use of the term CI. First let me point out that I meant for the inclusion symbols in Figure 1 to be taken quite literally. Note that I show CI as a SUBSET of AI. I believed this to be the case in 1992, and I still believe it in 2013, 21 years later (look right now at the second column of Table 2 to understand why I still hold this opinion). Second, you can see from the abstract (quote Q5) that my main focus was on CNNs – *computational neural networks* – and their relationship to AI, and more generally, BI. I was particularly concerned about the way many writers spoke about NNs. I wrote

Quote Q6: (Bezdek, 1992)

> Another objective concerns the use of "*seductive semantics*"; that is, words or phrases which convey, by being interpreted in their ordinary (non-scientific) usage, a far more profound and substantial meaning about the performance of

> an algorithm or computational architecture than can be readily ascertained from the available theoretical and/or empirical evidence. Examples of seductive phrases include words such as: *neural, self-organizing, machine learning, adaptive, cognitive.*

Here's an example of what I meant. There is an IOS journal titled *Intelligent Data Analysis*. What do you think the articles in it are about? Can you imagine doing **un-**intelligent data analysis, and asking anyone to publish your results? Of course not (but unintelligent papers get published anyway)! I retrieved this blog from a quick Internet search on the query "Buzzwords gone bad." A note from Jonathan Chizik dated August 9, 2011 says: "keynote speaker here just said one of his company's tactics to succeed is to "think intelligently"... as opposed to what, thinking stupidly?" To which, David McBride replied the next day: "They should proactively leverage their synergies by trying harder to think intelligently." Now do you see what I meant? The phraseology and semantics of computation that I was attempting to capture and discuss was exhibited in Table 1 of the 1992 paper, repeated as Table 1 here. This table makes my ideas about what CI might mean pretty clear.

Table 1. Defining the ABC's (Bezdek, 1992, 1994)

BNN	Hardware: The *BRAIN processes your sensory inputs*	Sensory Data Processing; How *does it Work*?
ANN	CNN (+) "Knowledge Tidbits (KTs)" process sensor inputs and KTs *in the style of the brain*	Intermediate level processing. **More** than "adaptivity, fault tolerance", ... etc. A human is always in the loop.
CNN	Biologically inspired models process Sensor Inputs *in the style of the brain*	Low-level sensor data processing.
BPR	*Search for structure in sensory data*	We are really good at it: *How does it Work?*
APR	CPR (+) "Knowledge Tidbits"	Intermediate level data processing which utilizes Knowledge Tidbits (**More** than sensor data).
CPR	*Search for structure in sensor data*	Fuzzy, Statistical, Deterministic, Heuristic. This includes almost all NN procedures.
BI	Software: The *MIND*	Cognition, Memory, Action: *How does it Work?*
AI	CI (+) "Knowledge Tidbits"	Intermediate level cognition *in the style of the Mind*
CI	Low level information analysis	Low-level cognition *in the style of the Mind.*

I believe that my 1992 paper spurred the IEEE Neural Networks Council (NNC) into using the term CI for their world congress, and subsequently, as the name of the society itself. Thus, the IEEE NNC/CIS essentially turned the term CI from an operetta (after all, topical satire is often a feature of the operetta) into a full-blown Broadway smash musical. This is analogous to the role played by Dempster et al. (1977) for the term EM. The main difference between these two papers (events) is that my paper concentrated on the semantics of the term CI, while Dempster, Laird and Rubin not only altered the semantics of their field (by introducing the term EM), but also provided technical details and analyses for their intended use of the term.

Here is the email that I sent to Roy Nutter, Russ Eberhart, Pat Simpson, Bob Marks, and Toshio Fukuda on April 9, 1992 that broached this term with the IEEE neural networks council (NNC) for the first time:

Thu Apr 9 12 :33 :11 1992
To: rsn@ece.wvu.wvnet.edu, rce@rti.rti.org, xm8@sdcc12.UCSD.EDU, marks@b lake.u.washington.edu, and d43131a@nucc.nagoya- u.ac.jp
From: jbezdek@trivia.coginst.uwf.edu
Subject: NEW name of council
Status: R
I suggest the COMPUTATIONAL INTELLIGENCE COUNCIL, later to become the COMPUTATIONAL INTELLIGENCE SOCIETY.

This suggestion was accepted by the NNC executive committee (Excom), and two months later the name *IEEE World Congress on Intelligent Systems* was changed to The *IEEE World Congress on Computational Intelligence* by the NNC administrative committee (Adcom) at its meeting in Baltimore on June 7, 1992. The first WCCI, held in Orlando in 1994, combined the two major conferences of the NNC (neural networks, fuzzy systems), with a new one on evolutionary computation (EC). The scope of the NNC was modified in 1991 to include both fuzzy systems and evolutionary computation. The revised scope, adopted in 1991 (visit the IEEE CIS history website for the source of this quote):

Quote Q7: SCOPE of the IEEE NNC in 1991 (https://history.ieee-cis.sightworks.net/)

"The field of interest of the [Neural Network] Council and its activities and programs shall be the theory, design, application and development of biologically and linguistically motivated computational paradigms involving neural networks, including connectionist systems, genetic algorithms, evolutionary programming, fuzzy systems, and hybrid intelligent systems in which these paradigms are contained."

So, in 1992 the main focus of the NNC was neural networks, with fuzzy systems and evolutionary computation emerging as important newcomers (to the NNC). We will have a look at the evolution of the IEEE CIS in a bit, but first, let me make a few comments about the growth of the term CI in the next few years. Channeling the words of Meng and

Dyk, the term CI became "the theme of an overpowering symphony" in the period 1992-2000. The next section tracks what seems to be the turning point for this tune.

5. Accelerando: 1992-2000 – CI goes Viral

Following the name change to IEEE WCCI, and most particularly after the IEEE WCCI in Orlando in 1994, many people started asking – what is CI? And many more simply jumped on the bandwagon, and started calling it "their field," much the way a mathematician might say "differential geometry" in response to the question –"what field are you in?" Then the skies opened up, and definitions flooded the landscape. A number of writers began to supply their own interpretations as the heft of the term increased. Who else weighed in? The following references provide you with a representative, but by no means exhaustive, sample of comments about the "definition" of computational intelligence.

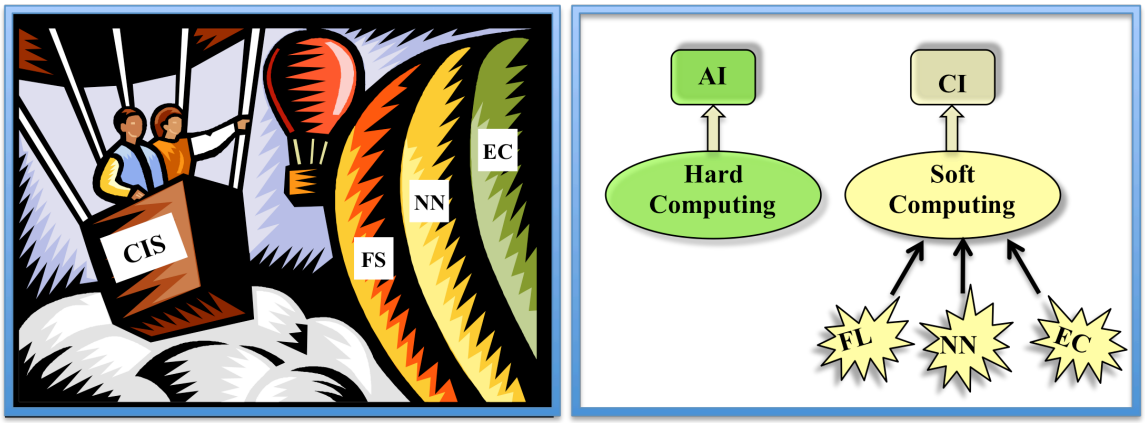
Bob Marks (1993) wrote an editorial summarizing his view of the differences between CI and AI, adding some interesting data about the disciplines that might be in CI and AI. He summarized his view this way: "Neural networks, genetic algorithms, fuzzy systems, evolutionary programming, and artificial life are the building blocks of CI." This was the so-called umbrella of CI (Figure 2a, the big 3) – the official party line of the IEEE NNC in 1993 for the basic disciplines comprising computational intelligence.

I wrote a second paper (Bezdek, 1994) for the book of invited papers associated with plenary lectures given at the 1994 WCCI. There is not much in it beyond Bezdek (1992). Eberhart (1995) offered his very different view of CI in 1995, and expanded on it in Eberhart et al. (1996), where we find "In summary, adaptation is arguably the most appropriate term for what computationally intelligent systems do. In fact, it is not too much of a stretch to say that computational intelligence and adaptation are synonymous." David Fogel critiqued my definition of CI in Fogel (1995), and argued that any definition that included the word intelligence necessitated discussion about and inclusion of the notion of (evolutionary) adaptation.

The structural organization of the IEEE NNC was delineated at the June 2, 1996 ADCOM meeting of the NNC by its then president, Walter Karplus, who affirmed that the scope of the IEEE NCC was still anchored by "the big 3," that is, the triumvirate of neural networks, fuzzy systems and evolutionary computation. He stated that "CI substitutes intensive computation for insight into how the system works. NNs, FSs and EC were all shunned by classical system and control theorists. CI umbrellas and unifies these and other revolutionary methods." This was perhaps the beginning of the umbrella paradigm that I schematized as Figure 3.1 in Bezdek (1998). I have redrawn that illustration here as Figure 2a, the hot air balloon being an appropriate update for the umbrella of a decade ago.

A new journal titled the *Int. J. of Computational Intelligence and Applications* (IJCIA) appeared in 2001. The statement of scope put forth at their website states, in part, that "The International Journal of Computational Intelligence and Applications, IJCIA, is a

refereed journal dedicated to the theory and applications of computational intelligence (artificial neural networks, fuzzy systems, evolutionary computation and hybrid systems)." This seems to reaffirm the big 3 party line espoused by the IEEE CIS about its aims and scope, seen in quotes Q7 above and Q13 below.



2a. Marks (1993); Karplus (1996)[4] **2b. Zadeh (1996)**

Figure 2. Definitions of CI (Figures 3.1 and 3.3, Bezdek, 1998)

Lotfi Zadeh offered another view about the meaning of and relationship between AI and CI to participants of the NATO advance study institute (ASI) held in Antalya, Turkey in the summer of 1996. A chart he presented there was reproduced in Bezdek (1998), replicated here as Figure 2b. At that time the term *soft computing* (SC) was also in its infancy, and Figure 2b provides us with an interesting interpretation of the relationship between the terms CI and SC, as well as a somewhat different view of the term AI. Zadeh did not show a direct connection between AI and CI as I did in Figure 1. Instead, he characterized the difference between AI and CI in terms of their underlying style of computation (hard vs. soft). This is really a very different interpretation of CI than any before it, and seems slanted more towards a defining view of soft computing than of CI or AI. The distinction between hard and soft computing *for me* is whether the *implemented models* are hard (crisp) or soft (fuzzy, probabilistic, possibilistic. I don't want to enter the fray about what SC is or is not, but to clarify Zadeh's chart, let me add that my own understanding of the two types of computing represented in Figure 2b is that the actual computing is always done in what I would call "the standard way."

It would be impossible, in 2013, to provide you with an accurate estimate of the numbers of laboratories, books, papers, journals, institutes, degree programs, and so on, that now use CI as if it's essential meaning was as well understood as a term such as, for example, "calculus." However, the discourse about what CI is (or is not) has not diminished. Indeed (and perhaps amusingly), Chapter 6 of the EOLSS volume titled *Artificial Intelligence* (Joost, 2009), authored by Craenen and Eiben (2009), has the title *Computational Intelligence*! This chapter has a pretty accurate and complete recounting

[4] Who are those two hardy balloonists? Frank Rosenblatt and Lotfi Zadeh? Nick Cercone and Gordon McCalla? Bart Kosko and Robert Hecht-Nielsen? Bob Marks and Walter Karplus? Mickey Mouse and Daffy Duck? You decide.

of the definitions of CI offered by Bezdek, Marks, Fogel and Eberhart. Here are some snippets of their discussion on this topic:

Quote Q8: (Craenen and Eiben, 2009)

> Although used fairly widespread, there is no commonly accepted definition of the term computational intelligence. Attempts to define, or at least to circumscribe, CI usually fall into one or more of the following categories:
>
> - Conceptual treatment of key notions and their roles in CI
> - "Relative definition" comparing CI to AI
> - Listing of the (established) areas that belong to it
>
> [...]
>
> **3. Artificial versus computational intelligence?**
>
> The relationship between computational intelligence and artificial intelligence has formed a frequently discussed issue during the development of CI. While the last quote from the previous section implies they are synonyms, the huge majority of AI/CI researchers concerned with the subject sees them as different areas, where either
>
> - CI forms an alternative to AI
> - AI subsumes CI
> - CI subsumes AI

Why has this term, and discussions about its presumptive meaning, become so popular? Well, channeling Meng and Van Dyk (1997), I could just as easily ask you why the song "White Christmas" is a monster hit that has been covered by hundreds of performers and continues to sell millions of records, while hoppin' and boppin' to "Jingle Bell Rock" simply gets relegated to some airtime on oldies stations every December? I suppose everyone has their own theory about this, and I want to conclude this section with a few comments about mine.

Let's return to the introductory paragraphs of the IJCI. The two sentences comprising the last paragraph in Quote Q4 are directly concerned with much of what I wanted to say in 1992 about Artificial Intelligence. "**AI is distinct from other studies of intelligence in its emphasis on computational models"**; and **"The name also seems short enough to be catchy.**" The first of these is related to the amount of computation done in AI – but note their choice of the word "emphasis," as opposed, for example, to a word such as "exclusively."

The second sentence concentrates on the importance of choosing a good *buzzword*. Before you dismiss this as cynicism or scorn, let me reassure you that I mean absolutely no disrespect to Nick Cercone and Gordon McCalla, or to the publishers of that journal. On the contrary, I admire their gumption and their foresight for admitting to their readers that "catchy" is important. Do a Google search on "buzzwords gone bad." You will find

page after page of links to articles about buzzwords in fashion design, government labs, resume writing, politics, Internet marketing, and on, and on, and on. Here's part of an article published in Marketing Today (2008) about the pervasive and destructive nature of buzzwords in business and industry:

Quote Q9: (BUZZWORDS GONE BAD [*Marketing Today*, Retrieved 2008-01-04.]

> "Companies claiming to create "synergies" in an effort to develop a "value-added" "paradigm" that leads to new "solutions" may want to be strategic in another way: not going overboard with cliché phrases and industry jargon. ... Buzzwords and industry jargon are a form of shorthand used by people within a particular company or profession, but they can be confusing or even seem exclusionary to individuals outside of that field," said Max Messmer, chairman of Accountemps and author of *Job Hunting For Dummies*® (John Wiley & Sons, Inc.). "When these words are overused, they can lose their impact altogether." Part of the motivation to use buzzwords can be attributed to a desire to demonstrate your expertise, but this can often backfire. Added Messmer, "Even though the terms you use may be clear to you, other people must understand them if you hope to communicate your point effectively. For instance, instead of saying a project was a 'win-win,' explain why it was successful." As society and pop culture evolve, old catchphrases die out, while new jargon is born.

Note the final sentence: *As society and pop culture evolve, old catchphrases die out, while new jargon is born*. Sound familiar? Data mining? Machine Learning/ Big Data? Q9 is a bit harsh – the implication is that buzzwords are bad. On the other hand, an editorial written by Mark Radford in October 2004, about the use of buzzwords in the software development industry has the following statement:

Quote Q10 (Radford, 2004)

> Perhaps we should learn from experience with TDD [Test-Driven Development] and take stock of practices that we would like to see adopted more widely, and then sharpen our skills in coming up with buzzwords and/or buzz-phrases that are sufficiently **catchy** for the majority of developers and/or managers. Then, as what happened to TDD happens to other useful practices, maybe the "Buzzword Adoption Pattern" will start to emerge.

There are two interesting points about this quote. First, Radford uses the word "catchy" (which I have made boldface for emphasis), just like its use in Q4 by the founding editors of IJCI. And second, Radford does not dismiss the buzzword as an annoying artifact of bad speaking and writing. Instead, he advocates an almost formal approach to the adoption of useful buzzwords. Apparently buzzwords come in two flavors: "good buzzwords" and "bad buzzwords." A quote from the website About.com/Grammar and Composition supports this explicitly:

Quote Q11 (About.com/Grammar and Composition, 2012)

> The Fortune 500 communications professionals surveyed for this stylebook are split down the middle when it comes to the use of **buzzwords** in business writing. Approximately half disdain buzzwords of any kind while the other half think some buzzwords are effective (for instance, *bottom line, globalize, incentivize, leverage, paradigm shift, proactive, robust, synergy* and *value-added*). As a general rule, use buzzwords judiciously, always keeping the readers in mind. "If a buzzword is lively and capable of injecting some spunk into a dull sentence (and it does not alienate the readers), then use it. (Cunningham and Greene, 2002)"

I believe that good buzzwords are an integral part of science and engineering! Here is the explanation I offered for the phenomenal growth of the term CI in 1998:

Quote Q12 [this includes footnote 2]: (Bezdek, 1998)

> Why? Well, I'm not really sure. But I suspect that there are two main reasons. First, the technical community is somewhat disenchanted with (perceptions, anyway, of) the basis of AI research. I will argue here that AI tackles really hard problems, and that goals may have been set unrealistically high in the early days of AI. And second, scientists and engineers have a certain hunger - maybe even a justifiable need - for new terms that will spark interest and help sell papers, grant proposals, research and development programs and even products. These are the defining characteristics of the so-called buzzword, of which CI is currently a prime example. After all, computational neural networks in their best-known form have been around since 1943 [10], evolutionary computation since 1954 [11][5], and fuzzy sets since 1965 [12]. Funding entities and journal editors get tired of the same old terms. Is my attitude about this a little too cynical? Probably. But I think it's pretty accurate.

Have things changed in 2013? I don't think so. It would be trivial for me to dismiss CI as just a "good" buzzword, but certainly that's a large part of its appeal and logical explanation for its astonishing growth. But when the opening sentence of a nomination statement for an IEEE award is "Prof. [deleted] has worked **in the area of computational intelligence** for over 35 years," it makes you wonder what people "outside" of our technical community should make of the statement. After all, if I tell you that Prof. X is an English professor, you may wonder "in what specialty – poetry, Yeats, science fiction, novellas, American literature, etc.?" but you will certainly have a pretty

---

[5] David Fogel, an ardent student and chronicler of the history of computational models that emulate evolution, told me "I think it would be a fair designation to say that evolutionary computation dates back to 1954 (Barricelli had a paper in the journal *Methodos* - it's in Italian - but the paper's intro is reprinted in a later paper in *Methodos* in 1957 that's in English)." History aficionados can do no better than chapter 3 of Fogel (1995b) for a more complete discussion.

good general idea about what Prof. X must do. Would you have the same reaction on encountering the sentence I just recounted about Prof. [deleted]? I am guessing no. Just what is the area of CI? Coming up.

6. Finale: CI in 2012

So, is CI more than just a buzzword? Yes and no: It is a high-level buzzword now. What does that mean? Well, 25 years ago, CI represented mostly NNs. Then FSs and EC came along to form the big 3 for the IEEE CIS. According to Table 2, CI has become a convenient catchphrase that covers a lot more ground. Let's start at the IEEE CIS website, which has the following definition of its scope:

Quote Q13: SCOPE of the IEEE CIS [IEEE CIS website, retrieved Feb. 1, 2012.]

> The Field of Interest of the Society shall be the theory, design, application, and development of biologically and linguistically motivated computational paradigms emphasizing neural networks, connectionist systems, genetic algorithms, evolutionary programming, fuzzy systems, and hybrid intelligent systems in which these paradigms are contained.

The topics identified in this statement of scope are not that far from the triumvirate of interests seen in Figure 2a that was first publicized as CI in 1993. Indeed, if you compare the scope for 2012 to that shown in Quote Q7 for 1991, the changes that seem to define the basic topics defining CI are mostly cosmetic. But if we retrieve the current list of publication activities of the IEEE CIS from their website, we find a much broader set of interests. From their website:

Quote Q14: PUBLICATIONS of the IEEE CIS [IEEE CIS website, retrieved Aug. 1, 2013.]

> We currently sponsor or co-sponsor ten *IEEE Transactions: Neural Networks and Learning Systems; Fuzzy Systems; Evolutionary Computation; Computational Biology and Bioinformatics; Autonomous Mental Development; Computational Intelligence and AI in Games, Nanobioscience, Information Forensics and Security*; *Affective Computing*; and *Smart Grids*.
>
> CIS also publishes the *Computational Intelligence Magazine* as a benefit of membership in CIS and sponsors an IEEE Press/Wiley book series on computational intelligence.

Judging from this list of current publications, the technical activities of the IEEE CIS reach far beyond its "official scope." Does this extended listing help us decide what the term CI means? For example, is "autonomous mental development" a subset of what people mean when they say "computational intelligence." Well, not for me, and probably not for many others. But it is interesting and provocative, for example, to see a title such

as the *IEEE Trans. in Computational Intelligence in AI and Games* under the IEEE CIS banner. This brings us full circle to the relationship between CI and AI, which I have already discussed at some length. We can get some additional perspective about this by looking at the tables of contents of the two EOLSS books titled *Artificial Intelligence* (Joost, 2009), and this volume, *Computational Intelligence* (Ishibuchi, 2013). Table 2 shows the tables of contents in adjacent lists.

Table 2. Tables of Contents of the EOLSS Books on AI and CI

***Artificial Intelligence* (Joost, 2009)**

1. Artificial Intelligence: Definitions, Trends, Techniques, and Cases
2. Logic in AI
3. Intelligent Agents
4. Dynamical Systems, Individual-based Modeling, Self-Organization
5. Machine Learning
6. **Computational Intelligence**
7. **Evolutionary Computation**
8. Quantum Computing
9. **Neural Networks**
10. **Fuzzy Logic**
11. DNA Computing
12. Knowledge Based System Development Tools
13. Speech Processing
14. Data Mining
15. Vision
16. Expert Systems

***Computational Intelligence* (Ishibuchi, 2013)**

1. History, Philosophy and Development of Computational Intelligence (this chapter)
2. History and Philosophy of Neural Networks
3. Recurrent Neural Networks
4. Adaptive Dynamic Programming and Reinforcement Learning
5. Associative Learning
6. Kernel Models and Support Vector Machines
7. History and Philosophy of Fuzzy Systems
8. Design and Tuning of Fuzzy Systems
9. Fuzzy Data Analysis
10. Type 2 Fuzzy Sets
11. Rough Sets
12. History and Philosophy of Evolutionary Computation
13. General Framework of Evolutionary Computation
14. Evolutionary Multiobjective Optimization
15. Memetic Algorithms and Hyper Heuristics
16. Swarm Intelligence
17. Artificial Immune Systems
18. Hybrid Computational Intelligence
19. Computational Intelligence and Medical Applications
20. Computational Intelligence and Smart Grid
21. Computational Intelligence and Computational Systems Biology
22. Computational Neuroscience
23. Neuromorphic Engineering
24. Brain-Machine Interface

Please notice that the three core topics identified by the IEEE CIS as CI are shown as chapters 7, 9, and 10 in the volume on AI. And further, Ch. 6 in the AI volume (Craenen and Eiben, 2009) is titled "computational intelligence." I have already commented on the Craenen and Eiben chapter. It's good. It accurately portrays part of the history of the term, and has a nice conclusion, that I will reproduce shortly. But the point here is that Ch. 6 of the AI volume appears to be on equal footing with Chs. 7, 9, and 10 of that book – to wit, CI, NN, FS and EC are all subfields of AI. This is not inconsistent with the view I put forth in Bezdek (1992) seen in Figure 1. I did not, in that 1992 paper, mention EC and FS as core technologies for CI, but since I offered CI in whatever manifestations that meant as a set of helper technologies for AI, I have no problem in seeing these topics in the TOC for the AI volume. Indeed, I think them both appropriate and well placed.

Craenen and Eiben suggest the addition of two more "core topics" to their interpretation of CI – viz., *DNA computing* (DNAC) and *quantum computing* (QC). They compare these five computational styles (FSC, NNC, EC, DNAC, QC) in this extended view from three vantage points: (i) the computational medium; (ii) parallelism; and (iii) inspiration from nature. In terms of the computational medium, they group {FSC, NNC, EC} as silicon-based computing, while {QC, DNAC} use different environments for the actual calculations. In a somewhat more controversial opinion, they call FSC an outlier to the other four styles in terms of parallelism. In terms of natural inspiration, they state that FSC and QC do not belong to "natural computation"- that is, having been inspired by natural processes. Again, I find this a bit puzzling, since the entire field of FS is based on the idea of representing natural language computationally. Puzzling, but so what? This is a good chapter to read, and provokes a lot of thought about what CI is and is not. I recommend it.

Now let's have a look at the topics on the right side of Table 2 – presumably the ones that define, in 2013, the disciplines comprising CI. In the main the topics shown still represent the big three. For example, I regard support vector machines as a subfield of CNNs, but you may wish to call it something else. Swarm intelligence? It's a low-level optimization technique. Some fields represented by publications under the IEEE CIS banner that don't have a sufficiently large current support set to warrant inclusion are missing here: virtual reality, financial engineering, autonomous mental development, game theory, bioinformatics, information forensics, etc. That's fine – topics come and go just like buzzwords, or cuisines.

I think the debate about "CI vs. AI," including questions such as "Is it CI or AI?" or "Does one of these areas include the other, do they overlap, etc. ?" are really pretty moot nowadays. I agree with Craenen and Eiben (2009), who remark that the boundary between CI and AI has diminishing borders. They point out the symbiosis of these two areas by noting that topics such as EC, FS and NN are frequently given a broad treatment in AI textbooks, while core publications such as the journal IJCIA consider symbolic AI as one of the areas integrated into CI.

CI has in fact become a fairly high-level term that encompasses lots of technical activities, much like the term "mathematics," or "physics," both of which can be divided

into finer and finer sets of more specialized areas. For example, just as mathematics ➟ differential equations ➟ ordinary differential equations ➟ linear ordinary differential equations, we can break down each branch of CI: evolutionary computation ➟ genetic algorithms ➟ mutation ➟ crossover, and so on. And in the vernacular of the day, when we say we "work in CI," it's a branding (of us) by this term. I find no harm in this, nor should I take offense if others disagree. After all, most of us don't care whether the Houston Astros baseball team plays their games in Astro Field, Enron Field, or Minute Maid Park, do we? It's still baseball; only the name of the playing field has changed over the years. For that matter, the Houston Astros began as the Colt 45s, but then Armstrong and Aldrin landed on the moon. And the rest, as they say, is history.

In the end, I don't think it's very important to categorize the meaning of this term anyway. The term itself has become a monster hit even if it doesn't suit your taste. And at the end of the day, what I say or think CI "means" is nothing more than my opinion. But I'm happy to have one, because as Herb Caen said many years ago, "any clod can have the facts – having an opinion is an art." So, now you have mine, and now the fat lady sings.

7. Glossary

Short form	Long form
Adcom	Administrative Committee
AI	Artificial Intelligence
aka	"also known as"
ANN	Artificial Neural Network
AO	alternating optimization
BI	Biological Intelligence
BNN	Biological Neural Network
CAIAC	Canadian Artificial Intelligence Association
CI	Computational Intelligence
CIS	(IEEE) Computational Intelligence Society
CNN	Computational Neural Network
CSCSUSCEIO	Canadian Society for the Computational Studies of Intelligence
DNAC	DNA computing
EC	Evolutionary Computation
EM	Expectation-Maximization (AO)
Excom	Executive Committee
FNN	fuzzy neural network (also NN computing)
FS	Fuzzy System
FSC	Fuzzy Systems Computing
GA	genetic algorithm
GAC	genetic algorithm computing
IEEE	Institute of Electrical and Electronics Engineers
IJCI	Int. J. Computational Intelligence
IJCIA	Int. J. Computational Intelligence and Applications
MLE	maximum likelihood estimation
MLP	multilayered perceptron

NN	Neural Network
NNC	(IEEE) Neural Networks Council
NRC	National Research Council
NRCC	National Research Council of Canada
QC	Quantum Computing
PR	Pattern Recognition
WCCI	World Congress on Computational Intelligence

8. Acknowledgment

David Fogel and Tim Havens reviewed this article. They found several errors of fact, and they also pounced on the many typographical errors I made. Beyond this, they added some great insights into what this article is all about. Thanks!

9. Bibliography

Bezdek, J. C. (1992). On the Relationship between Neural Networks, Pattern Recognition, and Intelligence, *Int. J. Approximate Reasoning*, 6(2), 85-107. (The first published definition of CI).

Bezdek, J. C. (1994). What is Computational Intelligence? *Computational Intelligence Imitating Life*, ed. J. Zurada, B. Marks and C. Robinson, IEEE Press, Piscataway,1-12. Reprinted in: DOE Proc. Adaptive Control Systems Tech. Symp. , ed. S. Biondo and C. J. Drummond, NTIS, Springfield, VA, 1995, 10-15. (An update to Bezdek (1992)).

Bezdek, J. C. (1998). Computational Intelligence Defined - by Everyone!, in *Computational Intelligence: Soft computing and Fuzzy-Neuro Integration with Applications*, eds. O. Kaynak, L. A. Zadeh, B. Turksen and I. J. Rudas, NATO ASI series F, v. 162, 10-37. (A review of other definitions of CI extant in 1998).

Cercone, N. (2012). (Personal email communication about the IJCI journal).

Cheeseman, P. (1988). An Inquiry into Computer Understanding, *Comp. Intell.*, 4, 57-142. (A rather savage attack on fuzzy modeling by Cheeseman, with replies and rejoinders by 22 authors – fun to read).

Craenen, B. C., Eiben, A. E. (2009). Computational Intelligence, Ch. 6 in *Artificial Intelligence*, [Ed. Joost Nico Kok], in Encyclopedia of Life Support Systems (EOLSS), Developed under the Auspices of the UNESCO, Eolss Publishers, Oxford ,UK. (A very nice summary and update of CI definitions, with additional opinions from the perspective of practicing AI-sters).

Cunningham, H. Greene, B. (2002). *The Business Style Handbook*. McGraw-Hill, NY, NY. (A nice text about business writing, with some good ideas about buzzwords).

Dempster, A. P., Laird, N. M. and Rubin, D. B. (1977) Maximum likelihood from incomplete data via the EM algorithm (with discussion). *J. R. Statist.* Soc. B, 39, 1-38. (As noted by Meng and Dyk, this paper established the term "EM" for AO to solve mixtures with MLE: the definitive cover that immortalized the mixture decomposition song).

Eberhart, R. (1995). Computational intelligence : a snapshot, in *Computational Intelligence : A Dynamic System Perspective*,, ed. Palaniswami, M., Attikiouzel, Y.,

Marks, R.J., Fogel, D. and Fukuda, T. , IEEE Press, Piscataway, NJ, 9-15. (Several definitions of CI are discussed).

Eberhart, R. , Dobbins, R. W. and Simpson, P.K. (1996). *Computational intelligence PC tools*, Academic Press Professional (APP), NY. (Extended discussions about what CI is and could become).

Fogel, D. (1995). Review of *Computational Intelligence Imitating Life*, ed. J. Zurada, B. Marks and C. Robinson, IEEE Press, Piscataway, NJ, *IEEE Trans. Neural Networks*, 6(6), 1562 - 1565. (Some interesting comments on the role of evolution in computational modeling).

Fogel, D. B.(1995). *Evolutionary Computation: Toward a New Philosophy of Machine Intelligence,* IEEE Press, Piscataway, NJ. (This is a great book to start you off on a safari into the jungle of evolutionary computation).

Fogel D.B. (1998) (ed). *Evolutionary Computation: The Fossil Record*, IEEE Press, NY, 1998.

Goldberg, D. E. (1989). Genetic Algorithms in Search, Optimization, and Machine *Learnin*g, Addison-Wesley, Reading, MA. (The cover of Gas that seemingly made it a major hit).

Holland, J. H. (1975). *Adaptation in Natural and Artificial Systems*, U. of Michigan Press, Ann Arbor, MI. (A classic introduction to evolutionary modeling).

IJCI (1985). First issue of the *Int. J. of Computational Intelligence*, Wiley-Blackwell, Feb. 1985. (I believe this to be the first published discussion of the term CI).

Ishibuchi, H. (2012). *Computational Intelligence*, in Encyclopedia of Life Support Systems (EOLSS), Developed under the Auspices of the UNESCO, Eolss Publishers, Oxford , UK. (The EOLSS book that contains this paper as Ch. 1).

Joost, Nico Kok (2009). *Artificial Intelligence*, in Encyclopedia of Life Support Systems (EOLSS), Developed under the Auspices of the UNESCO, Eolss Publishers, Oxford ,UK. (The EOLSS AI companion to this volume).

Marks, R. (1993). Intelligence : Computational versus Artificial, *IEEE Trans. Neural Networks*, 4(5), 737-739.

McCalla, G. (2012). (Personal email communication about the IJCI journal).

Meng, X.-L., Van Dyk, J. R. (1997). The EM Algorithm - An Old Folk song Sung to a Fast New Tune, *Jo. Royal Statist. Soc. B*, 59(3), 511-567. (A really interesting and comprehensive discussion of the origins, history, evolution and importance of the EM algorithm – and, fun to read too!)

Palaniswami, M., Attikiouzel, Y., Marks, R.J., Fogel, D. and Fukuda, T. (1995). Introduction to *Computational Intelligence : A Dynamic System Perspective*, ed. Palaniswami, M., Attikiouzel, Y., Marks, R.J., Fogel, D. and Fukuda, T., IEEE Press, Piscataway, NJ, 1-5. (The introduction contains some remarks towards the evolution of the term CI during its embroyonic state).

Radford, Mark (2004). Editorial: The Buzzword Adoption Pattern, ACCU article 243. (A pretty amusing piece that suggests finding a good buzzword algorithm).

Rumelhart, D. E. and McClelland, J. L. (1986). Parallel Distributed Processing, MIT Press, Cambridge, MA. (One of the classic texts on connectist systems).

Webster's New World Dictionary of Computer Terms (1988). 3rd ed., Prentice-Hall, Englewood Cliffs, NJ, 13.

Werbos, P. J. (1974). Beyond Regression: New Tools for Prediction and Analysis in the Behavioral Sciences. PhD thesis, Harvard University. (This thesis is the root of a lot of work on multi-layered CNNs. Werbos removed a roadblock to further developments in NNs by showing how to update the weights of more than one neuron at a time).

10. Biographical Sketch

Jim received the PhD in Applied Mathematics from Cornell University in 1973. Jim is past president of NAFIPS (North American Fuzzy Information Processing Society), IFSA (International Fuzzy Systems Association) and the IEEE CIS (Computational Intelligence Society, when it was the NNC): founding editor the *Int'l. Jo. Approximate Reasoning* and *the IEEE Transactions on Fuzzy Systems*: Life Fellow of the IEEE and IFSA; and a recipient of the IEEE 3rd Millennium, IEEE CIS Fuzzy Systems Pioneer, IEEE CIS Rosenblatt medals, and the IPMU Kampe de Feriet award. Jim's interests: woodworking, optimization, motorcycles, pattern recognition, cigars, clustering in very large data, fishing, co-clustering, blues music, visual clustering and poker. Jim retired in 2007, and will be coming to a university near you soon.

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